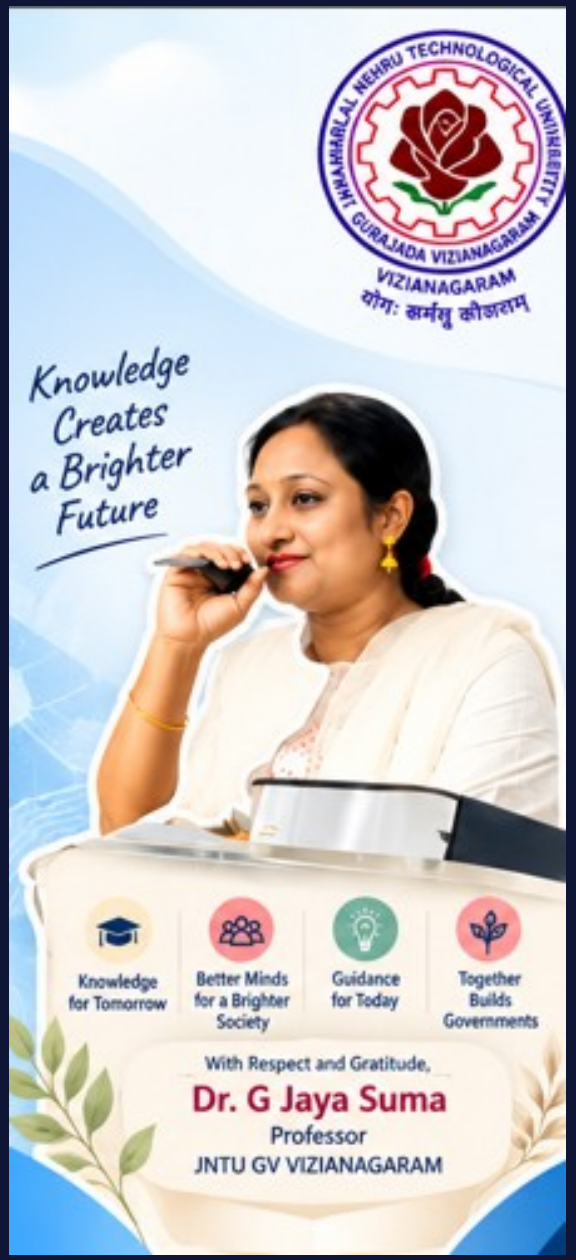


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VIDEO LECTURE SERIES · INTRODUCTION

From States to Operations

So far we described quantum states — now we learn how to change them, one operation at a time.

10 MIN

State | Operation | Reversible | Gates

1 What we'll cover in 10 minutes

Four short segments, from describing a state to changing one.

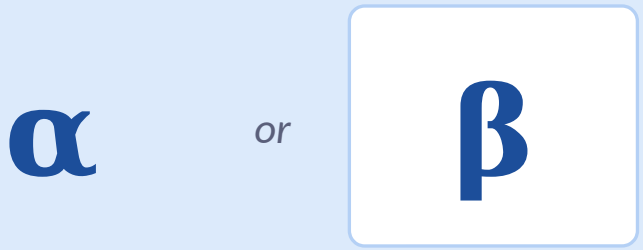
1. **Recap: what a state tells us** — 2 min
   Amplitudes, probabilities and measurement — everything so far describes a state.
2. **From describing to changing** — 3 min
   An operation takes one state and turns it into another.
3. **What an operation must preserve** — 3 min
   Total probability stays 1, and every step can be undone.
4. **Classical gates vs quantum operations** — 2 min
   Where the two ideas match, and where they differ.

2 Recap: what a state tells us

Before we change anything, let's recall what we can say about one qubit.

In simple words

So far we have described a qubit's state, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, and read it out by measuring. Now we ask how to change it.



The state is the complete description of the qubit; measuring only reveals part of it.

=

Described by amplitudes

Two complex numbers, α and β, with $|\alpha|^2 + |\beta|^2 = 1$.

◎

Read out by measuring

We get 0 or 1 with probabilities $|\alpha|^2$ and $|\beta|^2$ — and the state collapses.

?

Something is missing

Nothing so far lets us prepare or change a state on purpose — that is today's question.

3 From describing to changing

An operation takes one state and turns it into another.

An operation acts on a state

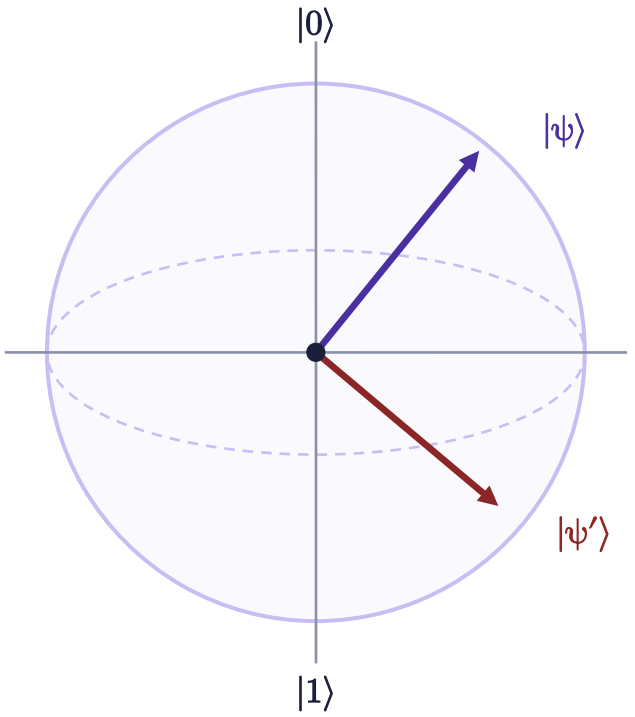
$$|\psi'\rangle = U\,|\psi\rangle$$

The operation **U** takes the state $|\psi\rangle$ and returns a new state $|\psi'\rangle$ — same qubit, different amplitudes.

$|\psi\rangle$ is what we start with; $|\psi'\rangle$ is what we would measure afterwards.

Key idea: an operation moves the state to a new point on the Bloch sphere — it is not a measurement.

On the Bloch sphere



The operation rotates the state vector; measurement is a separate, later step.

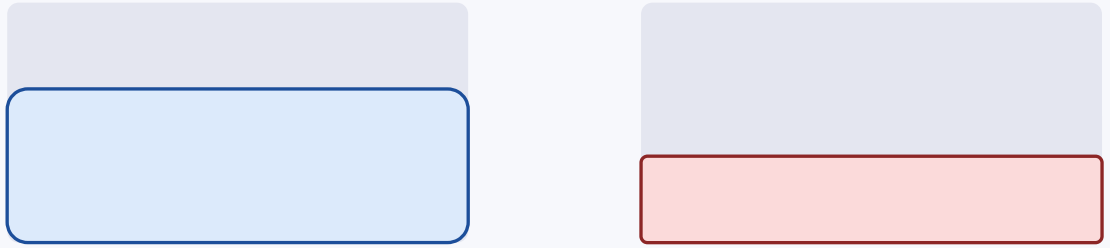
4 What an operation must preserve

Operations may move the state, but total probability stays 1 and nothing is lost.

Example: apply the X gate

$$0.6|0\rangle + 0.8|1\rangle \;\rightarrow\; 0.8|0\rangle + 0.6|1\rangle$$

X swaps the two amplitudes. The odds change from 36% / 64% to 64% / 36% — but they still add up to 100%.



After X: P(0) = 0.64

After X: P(1) = 0.36

Σ

Probability still sums to 1

0.64 + 0.36 = 1. An operation never creates or destroys probability.

↔

Always reversible

Apply X a second time and the original state returns — no information is lost.

U

Described by a matrix

Each operation is a unitary matrix — exactly the maps that keep total probability at 1.

5 Classical gates vs quantum operations

Same idea — input in, output out — but quantum operations follow stricter rules.

Property	Classical gate	Quantum operation
Acts on	A bit: 0 or 1	A state: amplitudes α and β
Reversible?	Often not (AND and OR lose information)	Always — every operation can be undone
Must preserve	Nothing in particular	Total probability equal to 1
Described by	A truth table	A unitary matrix
Creates superposition?	No	Yes — this is what gates like H do

ONE-LINE TAKEAWAY

An operation turns one state into another — reversibly, and without changing the total probability.

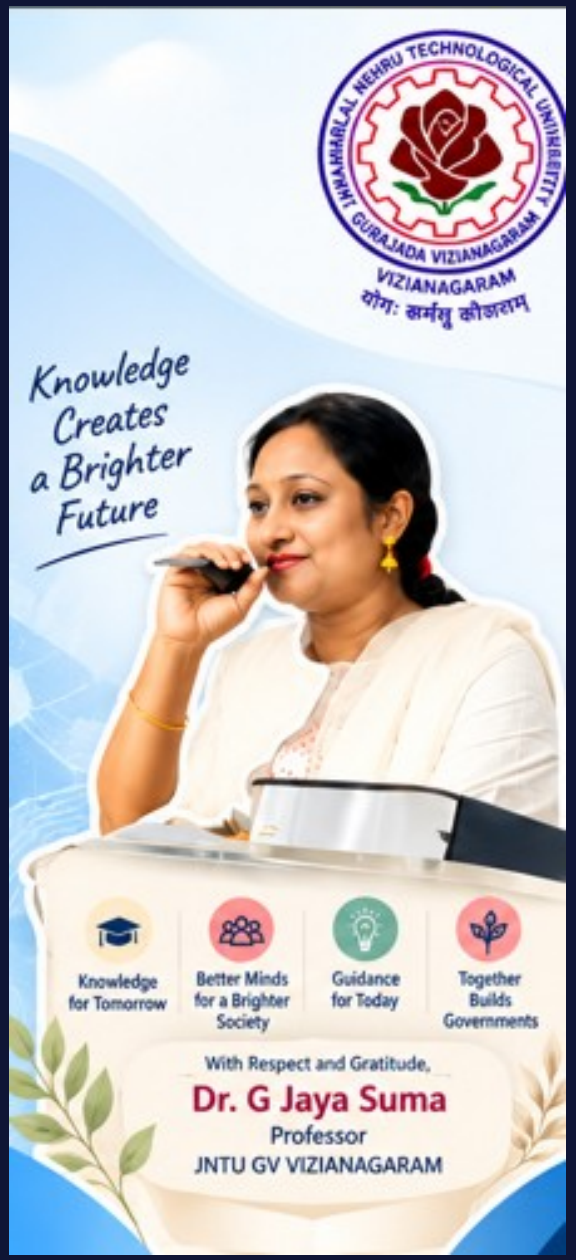
State in, state out

Always reversible

Probability stays 1

Next: Quantum Gates

We meet the first concrete operations — X, Z and H — and see exactly how each one moves a qubit.



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VIDEO LECTURE SERIES · REVERSIBLE COMPUTING

Reversible Computing & the Unitary Requirement

Why every quantum gate must be undoable — and what that demands of the mathematics.

35 MIN

Reversible | Landauer | Unitary | Toffoli

Fundamentals of Quantum Computing · Reversible Computing

1 What we'll cover in 35 minutes

Five segments, from why information loss matters to a fully reversible AND.

1. **Recap: why gates lose information** — 5 min
   Irreversible gates like AND throw information away — and that has a cost.
2. **The unitary requirement** — 8 min
   Why quantum gates must satisfy U†U = I, and what that means geometrically.
3. **Worked example: a unitary at work** — 7 min
   Apply a gate to a state, and see what a non-unitary map would break.
4. **Reversible gates: classical vs quantum** — 8 min
   NOT, CNOT and Toffoli, and how they look as quantum gates.
5. **Reversible AND: the Toffoli trick** — 7 min
   Keep the inputs, add a spare bit, and nothing is ever erased.

2 Recap: why gates lose information

Start with an ordinary AND gate: many possible inputs, one output.

In simple words

A gate is reversible if we can always recover its input from its output. AND is not: three different inputs all give 0.



Three different inputs, one output — from a 0 alone we cannot tell which input it came from.

& AND loses information

Inputs 00, 01 and 10 all give 0, so the original inputs cannot be recovered.

kT Erasing costs energy

Landauer's principle: erasing one bit releases at least kT ln 2 of heat.

↔ Reversible means no erasure

If every step can be undone, no information is erased — so no erasure cost has to be paid.

3 The unitary requirement

A quantum gate must be undoable, and must keep total probability equal to 1.

The condition on a gate U

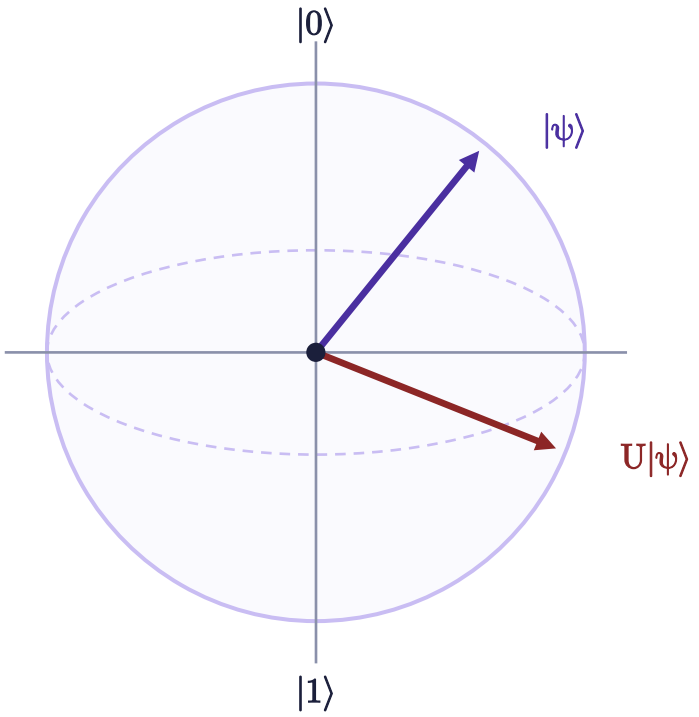
$$U^\dagger U = I$$

A gate **U is unitary** when U† (its conjugate transpose) is also its inverse: applying U, then U†, changes nothing.

Before: $|\alpha|^2 + |\beta|^2 = 1$. **After:** $|\alpha'|^2 + |\beta'|^2 = 1$ still.

Why it matters: a non-unitary map could push total probability above or below 1 — not a valid state.

On the Bloch sphere



A unitary rotates the state vector on the Bloch sphere — its length never changes.

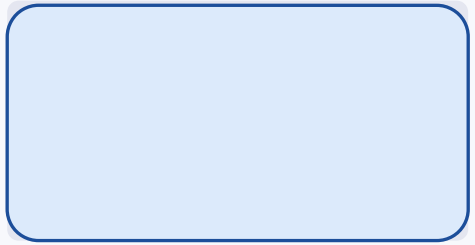
4 Worked example: a unitary at work

Apply the H gate to a state — and see what a non-unitary map would break.

Example: H applied to a state

$$H(0.6|0\rangle + 0.8|1\rangle) \approx 0.99|0\rangle - 0.14|1\rangle$$

The new amplitudes give P(0) = 0.98 and P(1) = 0.02 — they still add up to 100%, as a unitary guarantees.



After H: P(0) = 0.98



After H: P(1) = 0.02

Σ

Length preserved

0.98 + 0.02 = 1. A unitary never changes the total probability.

↔

H undoes itself

Apply H again and the original state comes back: 0.6|0⟩ + 0.8|1⟩.

≠

Non-unitary breaks it

A map sending both |0⟩ and |1⟩ to |0⟩ would give 1.4|0⟩: total probability 1.96, which is impossible.

5 Reversible gates: classical vs quantum

Reversible classical gates carry over directly; irreversible ones need a workaround.

Gate	Classical version	Quantum version
NOT	Flips a bit: 0 ↔ 1	X gate: swaps the $\|0\rangle$ and $\|1\rangle$ amplitudes
CNOT	Flips the target bit if the control is 1	Same action on basis states — and on superpositions
Toffoli (CCNOT)	Flips the target if both controls are 1	Same, as a 3-qubit unitary — universal for reversible logic
AND	Not reversible: 00, 01, 10 all give 0	No unitary version — use Toffoli with a spare bit
H	No classical version	Unitary, and creates superposition

6 Making AND reversible

Keep the inputs, add a spare bit, and nothing gets erased.

Before the gate

$|a, b, 0\rangle$

Two input bits, plus a spare (ancilla) bit that starts at 0.

↓ *Toffoli*

After the gate — nothing erased

a, b

inputs still kept

a AND b

now in the spare bit

Only the spare bit changes: it becomes a AND b.

No information lost

The output $|a, b, a \text{ AND } b\rangle$ shows the inputs exactly — so the gate can be reversed.

Spare bits cost space

Every ancilla bit has to be carried along, so reversible circuits usually need extra bits.

Uncompute to clean up

After using the result, run the gate again to reset the spare bit back to 0.

7 Toffoli in action: a reversible AND

With the third bit at 0, it ends up holding a AND b — and every output is different.

Toffoli flips the third bit only when the first two bits are both 1

Case 1	Case 2	Case 3	Case 4	Case 5: undo
Input (a b c)	Input (a b c)	Input (a b c)	Input (a b c)	Input (a b c)
0 0 0	0 1 0	1 0 0	1 1 0	1 1 1
Output	Output	Output	Output	Output
0 0 0	0 1 0	1 0 0	1 1 1	1 1 0

The takeaway for this segment: Cases 4 and 5 undo each other, so the gate is reversible. The spare bit holds a AND b, and no input information was erased.

ONE-LINE TAKEAWAY

Quantum gates must be reversible and probability-preserving — and that is exactly what being unitary means.

Irreversible gates erase bits

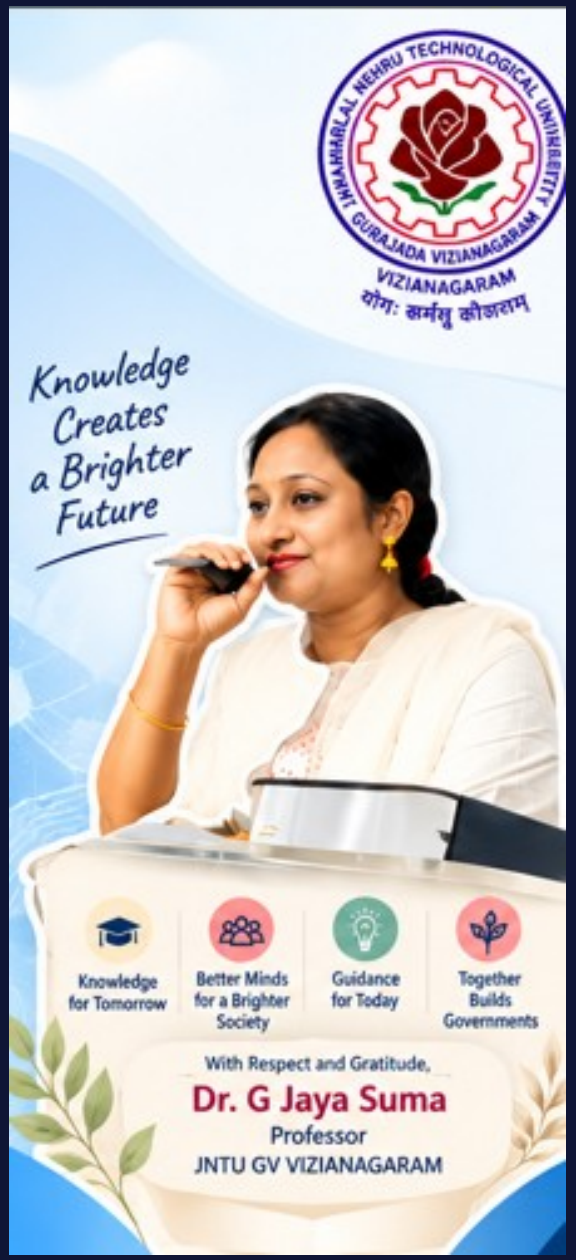
Unitary means U†U = I

Toffoli makes AND reversible

→

Next: Building circuits from gates

We take these unitary building blocks and combine them into circuits that compute something useful.



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VIDEO LECTURE SERIES · SINGLE-QUBIT GATES I

Single-Qubit Gates I: The Pauli Family

Meet X, Y and Z — three half-turns of the Bloch sphere that every other gate builds on.

40 MIN

Bit flip | Phase flip | Bloch sphere | Matrices

Fundamentals of Quantum Computing · Pauli Gates

1 What we'll cover in 40 minutes

Five segments, from gates as matrices to how X and Z move key states.

1. **Recap: gates as unitary matrices** — 5 min
   A single-qubit gate is a 2×2 unitary acting on the amplitudes.
2. **The X gate: the quantum NOT** — 9 min
   Swapping $|0\rangle$ and $|1\rangle$, and what that looks like on the Bloch sphere.
3. **The Z gate: a phase flip** — 8 min
   Changing a sign that a 0/1 measurement cannot see.
4. **The Pauli family at a glance** — 9 min
   I, X, Y and Z side by side, with their matrices.
5. **Pauli identities & key states** — 9 min
   Self-inverse gates, anticommutation, and how X and Z move key states.

2 Recap: gates are unitary matrices

A single-qubit gate is a 2×2 unitary matrix acting on the amplitude pair (α, β).

In simple words

A single-qubit gate is a 2 × 2 unitary matrix. Multiply it by the column of amplitudes (α, β) to get the new amplitudes.

2×2 *acts on* **(α, β)**

Unitary: the matrix can be undone and keeps total probability at 1.

U

Unitary

The rows and columns are orthonormal, so U†U = I.

↔

Reversible

Every gate has an inverse gate that undoes it exactly.

σ

Three special gates

Today: X, Y and Z, plus the identity I — together, the Pauli family.

3 The X gate: the quantum NOT

It swaps $|0\rangle$ and $|1\rangle$ — a half-turn of the Bloch sphere about the x-axis.

What X does to basis states

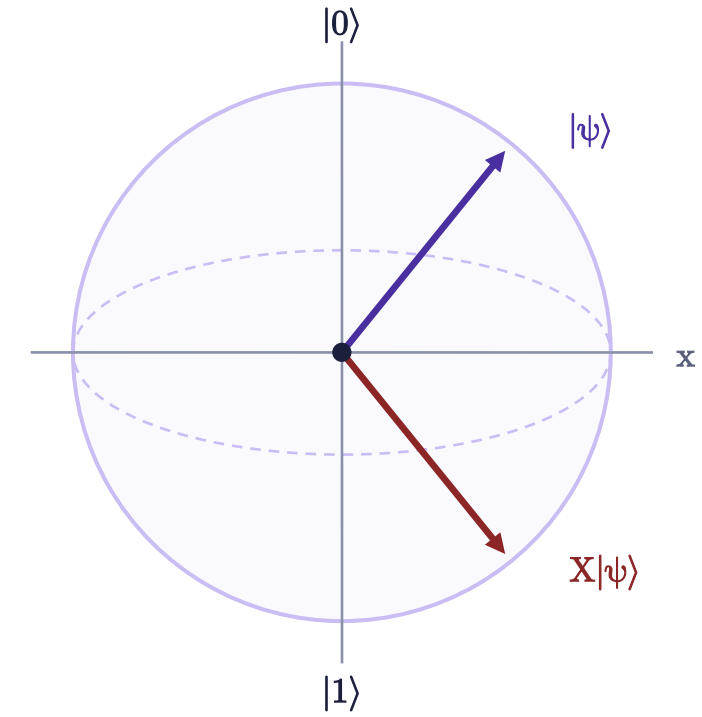
$$X|0\rangle = |1\rangle \;,\quad X|1\rangle = |0\rangle$$

X is the quantum **NOT gate**: it swaps the two amplitudes, so $\alpha|0\rangle + \beta|1\rangle$ becomes $\beta|0\rangle + \alpha|1\rangle$.

Matrix: X = [[0, 1], [1, 0]] — the identity with its columns swapped.

Watch out: X flips $|0\rangle$ and $|1\rangle$, but it leaves $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ unchanged.

On the Bloch sphere



A half-turn about the x-axis: the north pole goes to the south pole and the tilted state flips over.

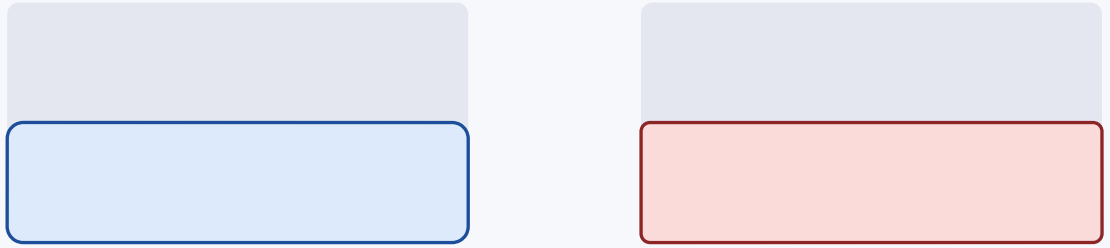
4 The Z gate: a phase flip

Z leaves $|0\rangle$ alone and flips the sign of $|1\rangle$ — a change a 0/1 measurement cannot see.

Example: Z applied to $|+\rangle$

$$Z\,|+\rangle = |-\rangle$$

Z flips the sign of the $|1\rangle$ amplitude, turning $|+\rangle$ into $|-\rangle$ — the amplitudes keep the same size, 0.707 each.



After Z: P(0) = 0.5

After Z: P(1) = 0.5

=

Same probabilities

Both outcomes are still 50/50 — a 0/1 measurement cannot see the flip.

φ

Only the phase changed

The relative sign between the amplitudes is new — a phase, not a probability.

H

Visible after H

Apply H to $|-\rangle$ and you get $|1\rangle$ with certainty, so the sign can be detected.

5 The Pauli family at a glance

Four 2×2 unitary matrices: the identity, and three half-turns of the Bloch sphere.

Gate	Matrix	What it does
I (identity)	[[1, 0], [0, 1]]	Does nothing to the state
X	[[0, 1], [1, 0]]	Bit flip: $\|0\rangle \leftrightarrow \|1\rangle$ (half-turn about x)
Y	[[0, −i], [i, 0]]	Bit and phase flip (half-turn about y)
Z	[[1, 0], [0, −1]]	Phase flip: $\|1\rangle \to -\|1\rangle$ (half-turn about z)
Products	XY = iZ, YZ = iX, ZX = iY	Any two Paulis multiply to the third, up to a factor of ±i

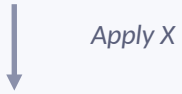
6 Applying a Pauli gate twice

Each Pauli gate undoes itself — and their order matters only by a sign.

Start

$$|\psi\rangle = 0.6\ |0\rangle + 0.8\ |1\rangle$$

Apply X once, then apply X again, and watch the amplitudes.



Result after each application

$0.8|0\rangle + 0.6|1\rangle$

after X once

$0.6|0\rangle + 0.8|1\rangle$

after X twice: the start

The same holds for Y and Z: each Pauli gate undoes itself.

$P^2 = I$

X, Y and Z each square to the identity, so applying any of them twice does nothing.

Hermitian and unitary

Each Pauli matrix equals its own conjugate transpose, so $P\dagger = P = P^{-1}$.

Order matters — by a sign

$XY = -YX$: swapping the order of two different Paulis only changes the sign.

7 X and Z on key states

Each gate is a half-turn of the Bloch sphere, so some states stay put and others swap.

Results are shown up to a global phase, which has no physical effect

$\|0\rangle$	$\|1\rangle$	$\|+\rangle$	$\|-\rangle$	$\|+i\rangle$
After X: $\|1\rangle$	After X: $\|0\rangle$	After X: $\|+\rangle$	After X: $\|-\rangle$	After X: $\|-i\rangle$
After Z: $\|0\rangle$	After Z: $\|1\rangle$	After Z: $\|-\rangle$	After Z: $\|+\rangle$	After Z: $\|-i\rangle$

The takeaway for this segment: X swaps the poles and fixes $|+\rangle$; Z fixes the poles and swaps $|+\rangle$ with $|-\rangle$ — each is a half-turn about its own axis.

ONE-LINE TAKEAWAY

The Pauli gates X, Y and Z are half-turns of the Bloch sphere — each is unitary, and each undoes itself.

X flips the bit

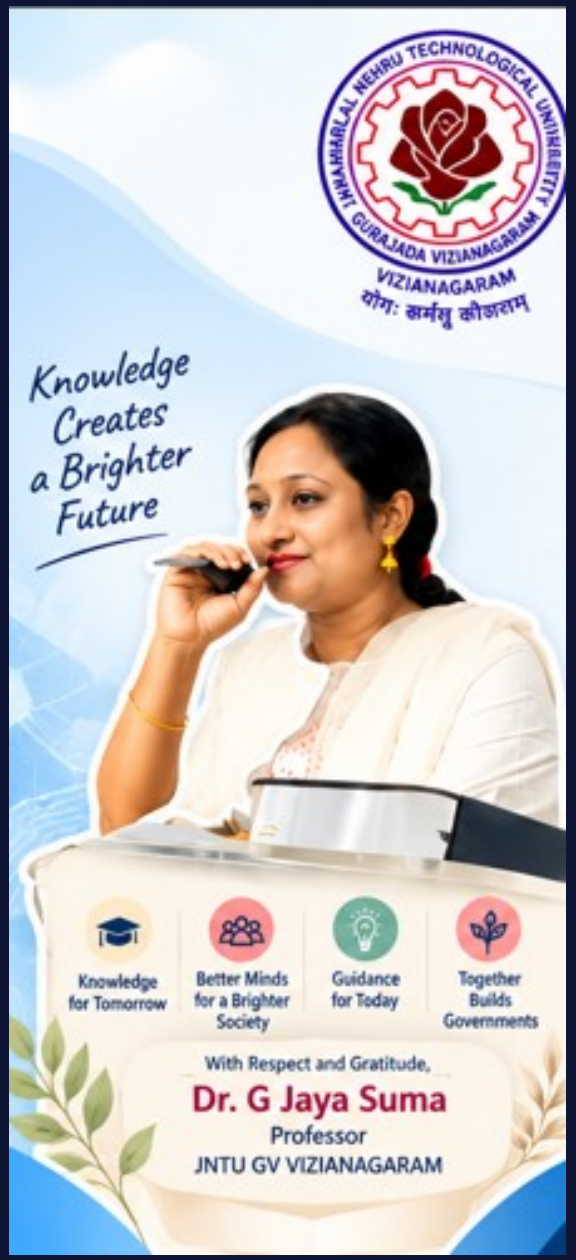
Z flips the phase

Y does both

→

Next: Single-Qubit Gates II

We move beyond half-turns to gates like H, S and T that rotate the state by other angles.



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VIDEO LECTURE SERIES · SINGLE-QUBIT GATES II

Single-Qubit Gates II: Hadamard, Phase and T

Create superposition with H, then steer the phase with S and T.

40 MIN

Hadamard | Phase | T gate | Rotations

Fundamentals of Quantum Computing · Hadamard & Phase Gates

1 What we'll cover in 40 minutes

Five segments, from what Pauli gates cannot do to how H, S and T fit together.

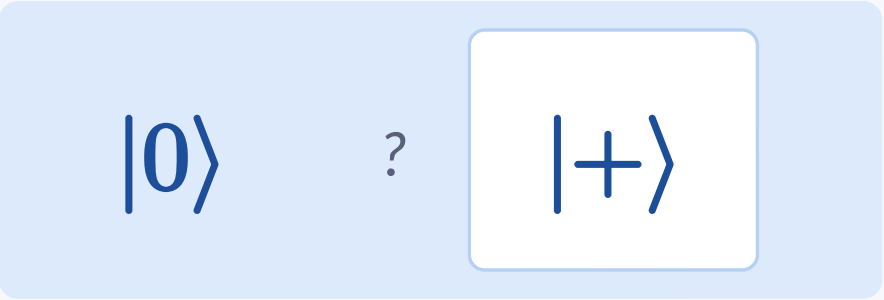
1. **Recap: Pauli gates as half-turns** — 5 min
   X, Y and Z flip states — but none of them creates superposition.
2. **The Hadamard gate** — 9 min
   Turning basis states into equal superpositions, and back again.
3. **Phase gates S and T** — 8 min
   Phases that hide from 0/1 measurements, and how H reveals them.
4. **H, S and T at a glance** — 9 min
   Matrices, actions and the identities that link them.
5. **Chaining gates and phase steps** — 9 min
   Why the order of gates matters, and how T, S and Z fit together.

2 Recap: Pauli gates as half-turns

X, Y and Z flip bits or signs — but none of them can create a superposition from |0⟩.

In simple words

The Pauli gates turn the Bloch sphere by half a turn. They flip bits or signs, but they never turn |0⟩ into an equal superposition.



No Pauli gate maps |0⟩ to |+⟩ — we need a new gate, and its name is Hadamard.

X

X flips the bit

X swaps |0⟩ and |1⟩ — a half-turn about the x-axis.

Z

Z flips the phase

Z turns |+⟩ into |−⟩ — a half-turn about the z-axis.

?

A gap remains

No Pauli gate makes superposition from |0⟩, and none offers finer phase steps.

3 The Hadamard gate: making superposition

H turns a basis state into an equal superposition — and back again.

What H does to basis states

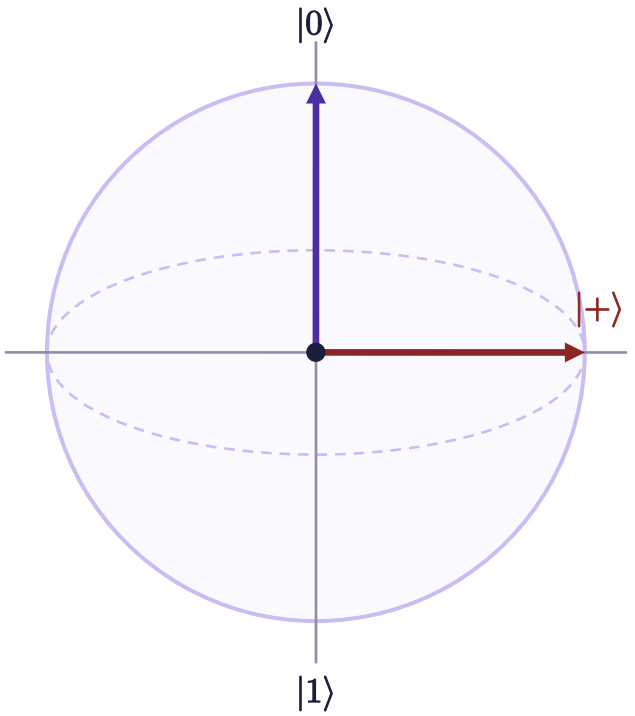
$$H|0\rangle = |+\rangle \;,\quad H|1\rangle = |-\rangle$$

H is the **superposition gate**: $|+\rangle$ and $|-\rangle$ both have amplitudes of size 0.707, so a measurement gives 0 or 1 with 50% each.

Matrix: H = 0.707 × [[1, 1], [1, −1]] — and applying H twice gives the identity.

Geometry: H is a half-turn about the axis halfway between x and z.

On the Bloch sphere



The north pole $|0\rangle$ lands on the x-axis at $|+\rangle$ — the equator, where 0 and 1 are equally likely.

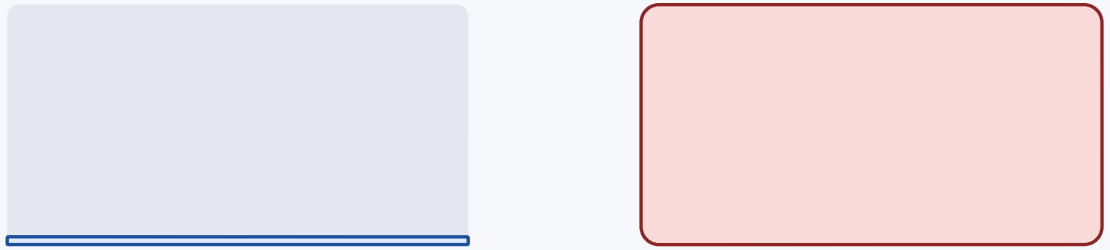
4 Phase gates: hidden until H

S and T change only the phase — but between two H gates, phase decides the outcome.

Example: H, S, S, H applied to $|0\rangle$

$$H\,S\,S\,H\,|0\rangle = |1\rangle$$

Two S gates make Z, and H Z H = X — so a hidden phase turns $|0\rangle$ into a certain $|1\rangle$.



i

S: a quarter-turn

S multiplies the $|1\rangle$ amplitude by i — a quarter-turn about the z-axis.

T

T: an eighth-turn

T multiplies $|1\rangle$ by $(1 + i)/\sqrt{2}$ — an eighth-turn about z, and $T^2 = S$.

H

Phase shows up after H

Between two H gates, a change of phase becomes a change in the 0/1 probabilities.

5 H, S and T at a glance

Three gates, three matrices — and a few identities that tie them together.

Gate	Matrix	What it does
H (Hadamard)	$0.707 \times [[1, 1], [1, -1]]$	$\|0\rangle \to \|+\rangle$ and $\|1\rangle \to \|-\rangle$ (superposition)
S (phase)	$[[1, 0], [0, i]]$	Multiplies $\|1\rangle$ by i: quarter-turn about z
T	$[[1, 0], [0, (1 + i)/\sqrt{2}]]$	Multiplies $\|1\rangle$ by $(1 + i)/\sqrt{2}$: eighth-turn about z
Stacking phases	$T^2 = S$, $S^2 = Z$	Each gate is two steps of the one before
Conjugating by H	$HZH = X$, $HXH = Z$	H swaps the roles of the x- and z-axes

6 Chaining gates: order matters

The same two gates in a different order can give a different state.

Start

$$|\psi\rangle = |0\rangle$$

Apply H and S to the same starting state, in both orders, and compare.

H and S

The result depends on the order

$|+i\rangle$

H first, then S

$|+\rangle$

S first, then H

As matrices, gates act right to left: S H means H acts first.

Read right to left

In S H$|\psi\rangle$, the gate written closest to the state, H, is applied first.

Different order, different state

H then S gives $|+i\rangle$, while S then H gives $|+\rangle$ — both valid, but not equal.

Gates rarely commute

Only special pairs give the same result in either order; most pairs do not.

7 Phase gates are z-rotations

Each T adds an eighth-turn, so S, Z and the identity are all built from T.

Applying T repeatedly turns the $|1\rangle$ amplitude by a growing phase angle and leaves $|0\rangle$ untouched

1 × T	2 × T	3 × T	4 × T	8 × T
Phase on $\|1\rangle$	Phase on $\|1\rangle$	Phase on $\|1\rangle$	Phase on $\|1\rangle$	Phase on $\|1\rangle$
45°	90°	135°	180°	360°
Same as	Same as	Same as	Same as	Same as
T	S	S·T	Z	Identity

The takeaway for this segment: $T^2 = S$, $T^4 = Z$ and $T^8 = I$. Circuits of H, S and CNOT alone can be simulated classically; adding T gives a universal gate set.

ONE-LINE TAKEAWAY

H makes superposition, S and T steer the phase — and together, H and T can approximate any single-qubit gate.

H: superposition

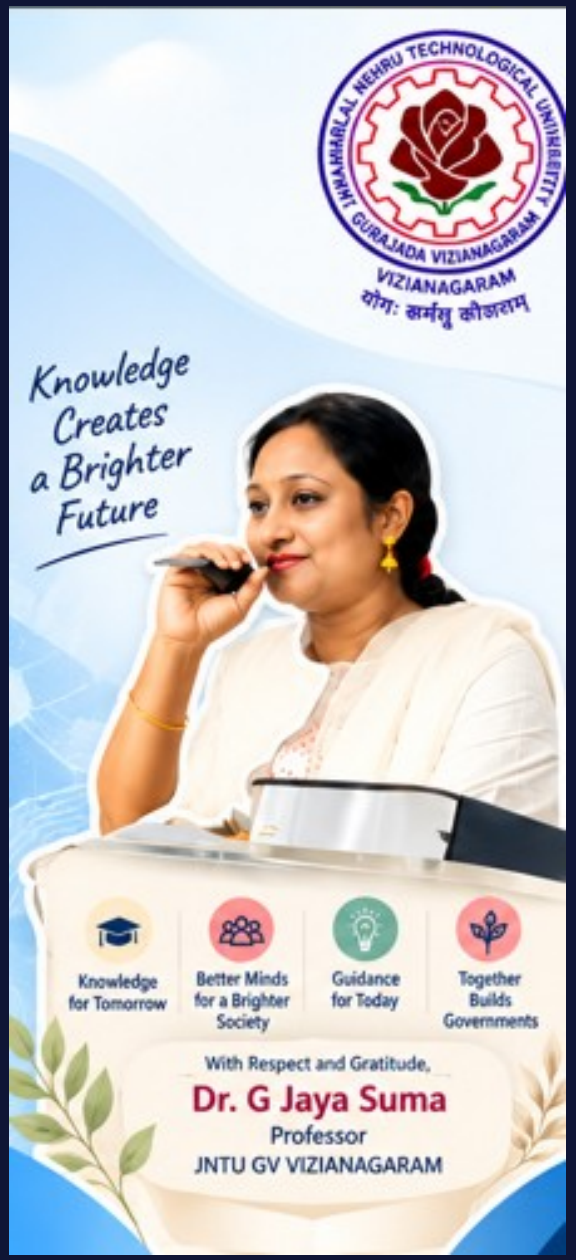
S and T: phase steps

Order of gates matters

→

Next: Two-Qubit Gates

We add a second qubit and meet CNOT — the gate that creates entanglement.



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VIDEO LECTURE SERIES · MULTI-QUBIT GATES

Multi-Qubit Gates

How gates act on two or three qubits at once — and how CNOT creates entanglement.

45 MIN

CNOT | CZ | SWAP | Toffoli

1 What we'll cover in 45 minutes

Five segments, from two-qubit states to controlled gates in general.

1. **Recap: from one qubit to two** — 5 min
   Four amplitudes, four basis states, and 4×4 unitary matrices.
2. **The CNOT gate** — 10 min
   Flip the target qubit only when the control qubit is 1.
3. **Worked example: making a Bell state** — 10 min
   H then CNOT turns $|00\rangle$ into an entangled pair.
4. **Multi-qubit gates at a glance** — 10 min
   CNOT, CZ, SWAP and Toffoli side by side.
5. **Controlled gates in general** — 10 min
   Any single-qubit gate can be controlled; CNOT and CZ are special cases.

2 Recap: from one qubit to two

Two qubits need four amplitudes — and gates that act on all of them at once.

In simple words

Two qubits have four basis states, each with its own amplitude. A two-qubit gate is a 4×4 unitary matrix acting on all four at once.

2 qubits → **4 states**

The state of two qubits is a list of four amplitudes whose squared sizes add up to 1.

4

Four amplitudes

One amplitude for each of $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.

4×4

Four-by-four gates

A two-qubit gate is a 4×4 unitary matrix acting on those four amplitudes.

⊗

Not always separable

Some two-qubit gates, like CNOT, cannot be written as two separate one-qubit gates.

3 The CNOT gate

Flip the target qubit only when the control qubit is 1.

What CNOT does to basis states

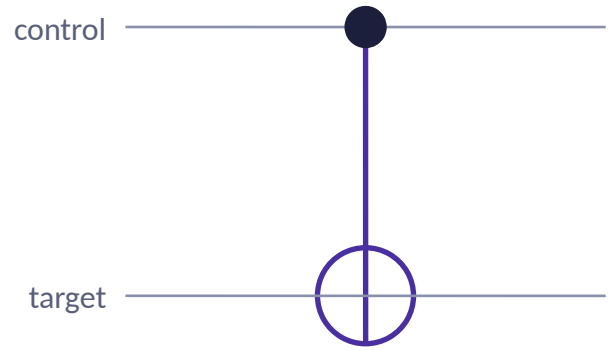
$$\text{CNOT}\ |a,\ b\rangle = |a,\ a \oplus b\rangle$$

CNOT flips the **target qubit** if and only if the **control qubit** is 1; the control itself never changes.

Control 0: target unchanged, so $|00\rangle$ and $|01\rangle$ stay put. **Control 1:** target flips, so $|10\rangle \leftrightarrow |11\rangle$.

Key idea: with a control in superposition, CNOT acts on both branches at once — that is how it entangles.

The CNOT circuit



$|0,\ b\rangle \rightarrow |0,\ b\rangle$

$|1,\ b\rangle \rightarrow |1,\ 1 \oplus b\rangle$

The dot is the control and ⊕ is the target; the target flips only when the control is 1.

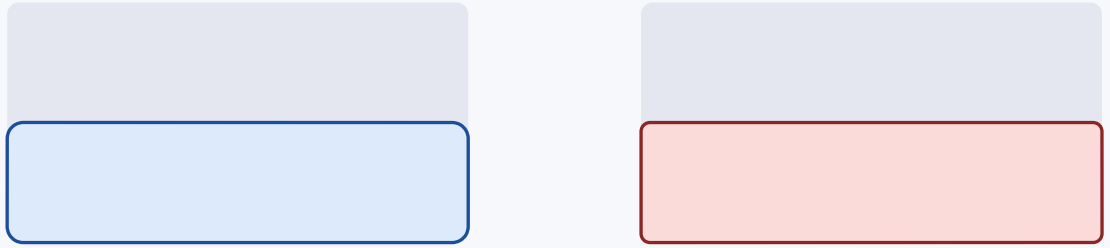
4 Worked example: making a Bell state

H on the first qubit, then CNOT: two separate qubits become an entangled pair.

Example: H, then CNOT, applied to $|00\rangle$

$$|00\rangle \rightarrow 0.707|00\rangle + 0.707|11\rangle$$

H gives $0.707|00\rangle + 0.707|10\rangle$; CNOT then flips the second qubit in the $|10\rangle$ branch, giving $|11\rangle$.



Not a product state

No pair of one-qubit states multiplies to give this state — the qubits cannot be described separately.

Perfect agreement

Each qubit alone gives a random 0 or 1, but the two results always match.

01 and 10 never occur

P(01) = P(10) = 0, so those two outcomes are never observed.

5 Multi-qubit gates at a glance

Four gates you will meet again and again, and how they act on basis states.

Gate	Action on basis states	What it is used for
CNOT	$\|a, b\rangle \to \|a, a \oplus b\rangle$	Creates entanglement; the workhorse two-qubit gate
CZ	$\|11\rangle \to -\|11\rangle$, all others unchanged	Flips the phase when both qubits are 1
SWAP	$\|a, b\rangle \to \|b, a\rangle$	Exchanges the states of two qubits
Toffoli (CCNOT)	Flips the target if both controls are 1	Three qubits: a reversible AND
Building blocks	SWAP = 3 CNOTs; CZ = H·CNOT·H on target	Each gate can be built from CNOT plus one-qubit gates

6 Controlled gates in general

Any single-qubit gate U can be controlled: it acts only when the control is 1.

Controlled-U

$$|c,\, t\rangle \rightarrow |c\rangle \otimes U^c|t\rangle$$

The control c is 0 or 1; U is applied once to the target t if c = 1, and not at all if c = 0.

Apply

Result for each value of c

c = 0

target unchanged

c = 1

U applied to target

With a control in superposition, both cases happen together.

CNOT = controlled-X

Take U = X and the target flips whenever the control is 1.

CZ = controlled-Z

Take U = Z and only $|11\rangle$ changes sign — CZ treats its two qubits symmetrically.

Toffoli = doubly controlled X

Two controls: X is applied to the target only when both are 1.

7 CNOT and CZ on basis states

CNOT swaps two basis states and CZ flips one sign — yet both can entangle superpositions.

States are written |control, target⟩

$\|00\rangle$	$\|01\rangle$	$\|10\rangle$	$\|11\rangle$	$\|{+}{+}\rangle$
After CNOT	**After CNOT**	**After CNOT**	**After CNOT**	**After CNOT**
$\|00\rangle$	$\|01\rangle$	$\|11\rangle$	$\|10\rangle$	$\|{+}{+}\rangle$ **unchanged**
After CZ	**After CZ**	**After CZ**	**After CZ**	**After CZ**
$\|00\rangle$	$\|01\rangle$	$\|10\rangle$	$-\|11\rangle$	**Entangled**

The takeaway for this segment: CNOT flips the target when the control is 1; CZ flips a sign when both qubits are 1. Both are unitary, and both can entangle.

ONE-LINE TAKEAWAY

Multi-qubit gates are unitary matrices on several qubits at once — and H plus CNOT is enough to entangle two qubits.

CNOT flips on control 1

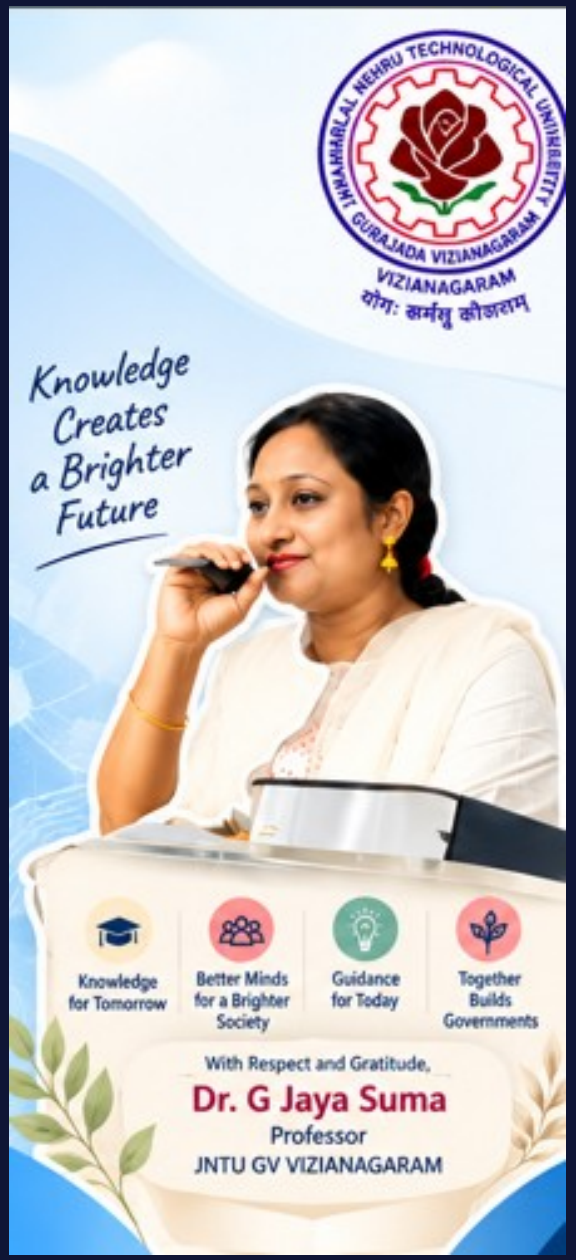
H + CNOT makes Bell states

Any gate can be controlled

→

Next: Quantum Circuits

We combine single- and multi-qubit gates into complete circuits and follow a computation from start to measurement.



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VIDEO LECTURE SERIES · QUANTUM CIRCUITS

The Quantum Circuit Model

Wires, gates and measurements — the standard way a quantum computation is written down.

40 MIN

Wires | Gates | Depth | Measure

1 What we'll cover in 40 minutes

Five segments, from gates as building blocks to the size of a whole circuit.

1. **Recap: gates as building blocks** — 5 min
   One-, two- and three-qubit gates are our raw material.
2. **Anatomy of a quantum circuit** — 9 min
   Wires, gates, time running left to right, and measurement at the end.
3. **Worked example: H, T, H** — 8 min
   Follow a three-gate circuit and see interference decide the answer.
4. **Circuit vocabulary at a glance** — 9 min
   Wire, gate, width, depth and measurement, side by side.
5. **From circuit to matrix** — 9 min
   Composing gates as a matrix product, and measuring a circuit's size.

2 Recap: gates as building blocks

We know gates on one, two and three qubits — the circuit model says how to combine them.

In simple words

So far we have met gates one at a time. A quantum circuit is a list of gates, applied in order, to a register of qubits.

Gates + **Order**

Circuit = the right gates, in the right order, on the right qubits.

1 **One-qubit gates**

X, Y, Z, H, S and T each rotate the state of a single qubit.

2 **Two-qubit gates**

CNOT and CZ can create entanglement; SWAP exchanges two qubits.

3 **Three-qubit gates**

Toffoli has two controls — a reversible AND.

3 Anatomy of a quantum circuit

Wires are qubits, boxes are gates, and time runs from left to right.

How to read a circuit

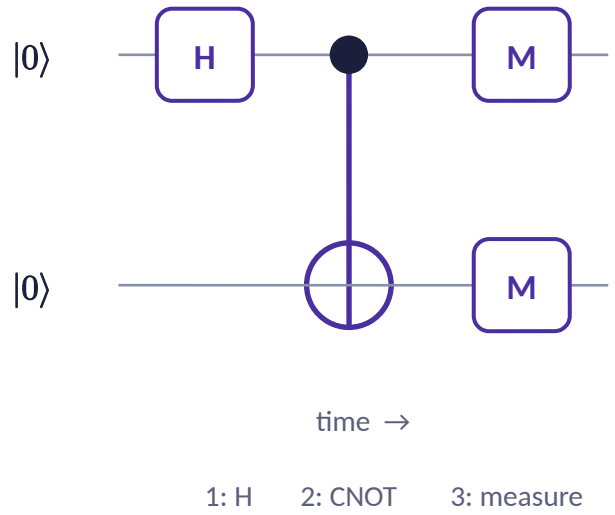
$$|00\rangle \rightarrow \mathrm{H} \rightarrow \mathrm{CNOT} \rightarrow \text{measure}$$

Each **horizontal line is a qubit**, boxes and dots are gates, and the M box at the end is a measurement.

Start: every qubit begins in $|0\rangle$. **End:** measuring gives one classical bit per qubit.

Key idea: the diagram is a recipe — the order of gates along each wire is the order in which they act.

A two-qubit circuit



Read left to right: H on the top qubit, then CNOT, then measure both.

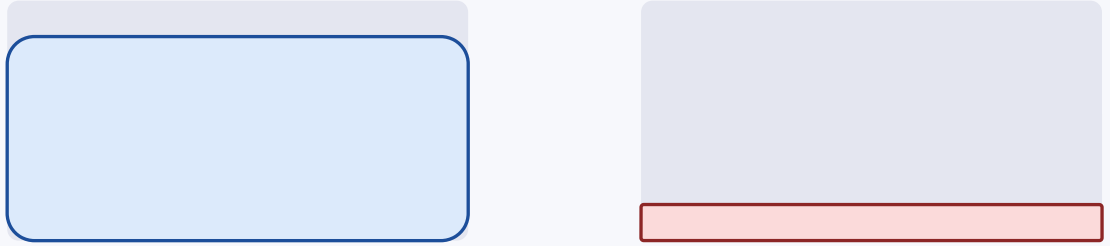
4 Worked example: H, T, H

Follow a three-gate circuit and watch interference decide the answer.

Example: a three-gate circuit

$$|0\rangle \rightarrow \mathrm{H} \rightarrow \mathrm{T} \rightarrow \mathrm{H} \rightarrow \text{measure}$$

H makes $|+\rangle$, T adds a 45° phase, and the second H turns that phase into probability: 0.85 for 0 and 0.15 for 1.



P(0) ≈ 0.85 P(1) ≈ 0.15

H

Interference

The second H turns a phase difference into a probability difference.

→

Order is time

Gates act in the order they appear, left to right — reorder them and the result can change.

Z

Swap one gate, change the answer

Replace T with Z and the circuit gives 1 every time: H Z H = X.

5

Circuit vocabulary at a glance

Five terms that describe almost every circuit diagram you will meet.

Term	Meaning	Example
Wire	One qubit's timeline, drawn left to right	q0 and q1 in the two-qubit circuit
Gate	A unitary operation on one or more wires	H on one wire; CNOT across two
Width	How many qubits the circuit uses	H then CNOT: width 2
Depth	How many layers of gates must run one after another	H then CNOT: depth 2
Measurement	Ends a wire, turning a qubit into a classical bit	The M box at the right end

6 From circuit to matrix

A whole circuit equals one matrix: the product of its gates.

Three gates in order

$$U = U_3\, U_2\, U_1$$

The first gate in time is the rightmost matrix, closest to the state it acts on.

Multiply

Two ways to write the same circuit

$U_1 \rightarrow U_2 \rightarrow U_3$

diagram: left to right

$U_3\, U_2\, U_1$

algebra: right to left

Both describe the same circuit; only the reading direction differs.

One big unitary

However long the circuit, it equals a single unitary matrix on all the qubits.

Undo it by reversing

Run the inverse gates in reverse order: $U\dagger = U_1\dagger\, U_2\dagger\, U_3\dagger$.

Measurement comes last

Unitary gates come first; the irreversible measurement is the final step.

7 Circuit size: width and depth

Width counts qubits; depth counts the layers of gates that must run one after another.

Width = number of wires · depth = number of sequential gate layers

Bell pair	GHZ state	H on all	SWAP	H T H
Width × depth	Width × depth	Width × depth	Width × depth	Width × depth
2 × 2	3 × 3	3 × 1	2 × 3	1 × 3
Gates used	Gates used	Gates used	Gates used	Gates used
H, CNOT	H, CNOT, CNOT	H, H, H	CNOT ×3	H, T, H

The takeaway for this segment: Every circuit is a list of unitary gates in order. Width tells us how many qubits we need, and depth tells us how many steps it takes.

ONE-LINE TAKEAWAY

A quantum circuit is a sequence of unitary gates on a register of qubits, ending in measurement.

Wires are qubits

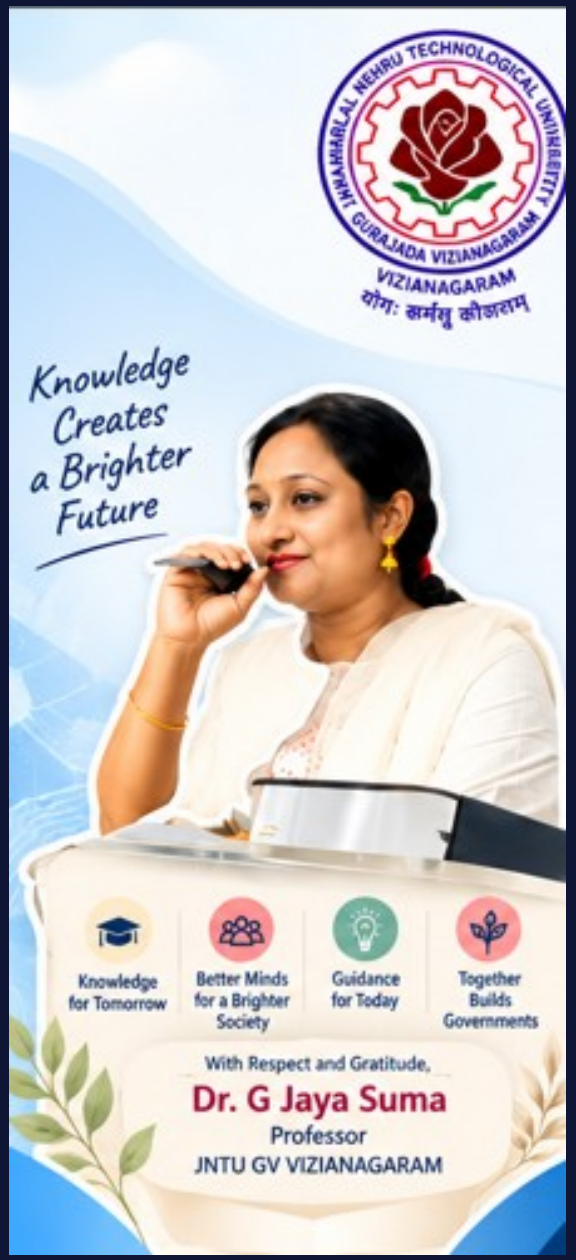
Gates act in order

Measure at the end

→

Next: Quantum Algorithms

We use circuits to solve a real problem, starting with the simplest quantum algorithms.



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VIDEO LECTURE SERIES · UNIVERSAL GATE SETS

Universal Gate Sets

Which small collections of gates are enough to build any quantum computation.

30 MIN

Exact | Approximate | Clifford | T gate

1 What we'll cover in 30 minutes

Five segments, from the idea of a gate library to how well a small set can do.

1. **Recap: from circuits to a gate library** — 4 min
   Circuits use gates from a fixed set — which small sets are enough?
2. **What universal means** — 6 min
   Why a finite gate set can only approximate every unitary.
3. **Worked example: X from H and T** — 7 min
   Build a Pauli gate out of two simple ingredients.
4. **Universal sets at a glance** — 6 min
   Which gate sets work, which do not, and why.
5. **Efficiency and exact constructions** — 7 min
   How fast approximation improves, and what we can build exactly.

2 Recap: from circuits to a gate library

A real device offers a short menu of gates — the question is which menu is enough.

In simple words

In classical logic, NAND alone can build every other gate. We ask the same question for quantum gates: is there a small set that builds everything?

NAND *builds* **all logic**

The quantum version of this question is what today's lecture answers.

N

Classical: NAND is enough

Every classical circuit can be built from NAND gates alone.

Q

Quantum: a bigger target

The goal is any unitary matrix on any number of qubits, not just a truth table.

?

A short menu

Real devices do a few gates well — so the menu must be short but powerful.

3 What universal means

A gate set is universal if its circuits can get as close as we like to any unitary.

The definition

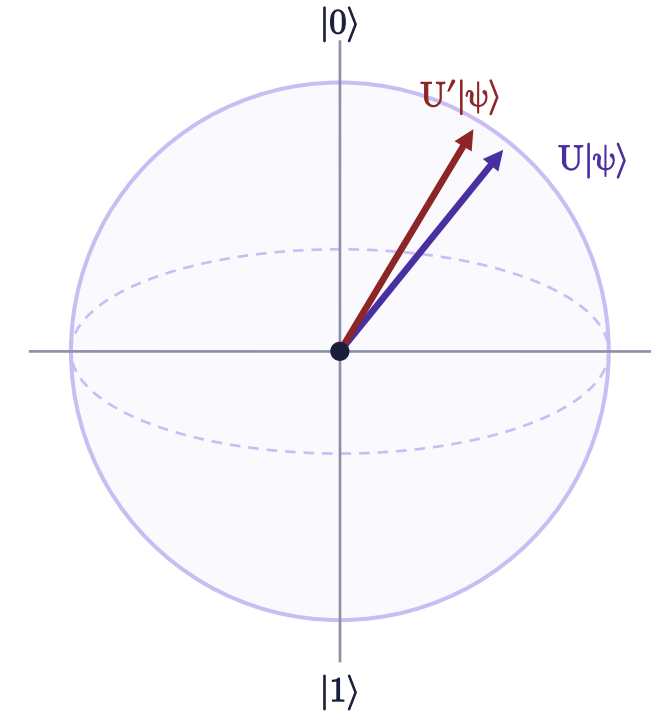
$$\|U - U'\| < \varepsilon$$

A gate set is **universal** if, for any target unitary U and any accuracy ε, some circuit U′ from the set satisfies the bound above.

Exact: copies U perfectly. **Approximate:** gets within ε of U — good enough.

Why approximate? a finite gate set builds only countably many circuits, but there are continuum-many unitaries.

On the Bloch sphere



The approximate state lands within a small angle of the target — measurement statistics barely change.

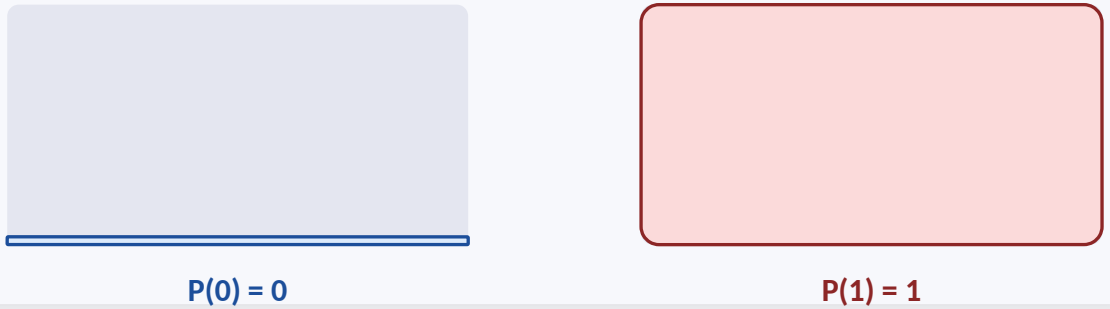
4 Worked example: X from H and T

Build a Pauli gate out of two simple ingredients — no approximation needed.

Example: an exact construction

$$X = H \cdot T^4 \cdot H$$

Four T gates make Z, and H Z H = X, so the circuit is exactly X. Applied to $|0\rangle$ it gives $|1\rangle$ with certainty.



S

T twice is S

$T^2 = S$: two T gates give a quarter-turn about z.

Z

S twice is Z

$S^2 = Z$, so four T gates give a half-turn about z.

H

H swaps X and Z

Conjugating by H turns Z into X — so H and T already reach every Pauli gate.

5 Universal sets at a glance

Some small collections work, and some fall short — for clear reasons.

Gate set	Universal?	Notes
{H, T, CNOT}	Yes — approximately	The standard Clifford+T set
{H, S, CNOT}	No	Clifford gates only: classically simulable
{Toffoli, H}	Yes — real amplitudes	Enough for quantum computation without complex phases
1-qubit gates + CNOT	Yes — exactly	Any unitary decomposes exactly, but needs arbitrary angles
{CNOT} alone	No	Never creates superposition

6 How fast does approximation improve?

The Solovay–Kitaev theorem: more accuracy costs only slightly more gates.

The guarantee

$$\text{gates} \approx [\log(1/\varepsilon)]^c$$

Ask for accuracy ε and a compiler finds a sequence of H and T gates whose length grows only polylogarithmically in 1/ε (c is a small constant).

Solovay–Kitaev

What it means in practice

ε	few gates
accuracy you ask for	length grows slowly

Halving the error costs only a small number of extra gates.

Slow growth

Much better accuracy needs only a modest number of additional gates.

A compiler's job

Software turns each rotation angle into a long H–T sequence automatically.

T gates are costly

On error-corrected hardware T gates are the expensive ones, so circuits count them.

7 Building familiar gates from a small set

Everything here is exact: built from H, T and CNOT with no approximation.

Constructions are equal up to a global phase, which has no physical effect

S	Z	X	Y	Toffoli
Built as	Built as	Built as	Built as	Built as
$T \cdot T$	$S \cdot S$	$H \cdot Z \cdot H$	$X \cdot Z$, up to phase	H, T, T†, CNOT
Why it works	Why it works	Why it works	Why it works	Why it works
$T^2 = S$	$S^2 = Z$	$HZH = X$	$Y = iXZ$	15 gates in all

The takeaway for this segment: Clifford gates and Toffoli can be built exactly from H, T and CNOT; arbitrary rotations cannot, and are approximated to accuracy ε.

ONE-LINE TAKEAWAY

A gate set is universal if its circuits can approximate any unitary — and H, T and CNOT are enough.

Approximate, not exact

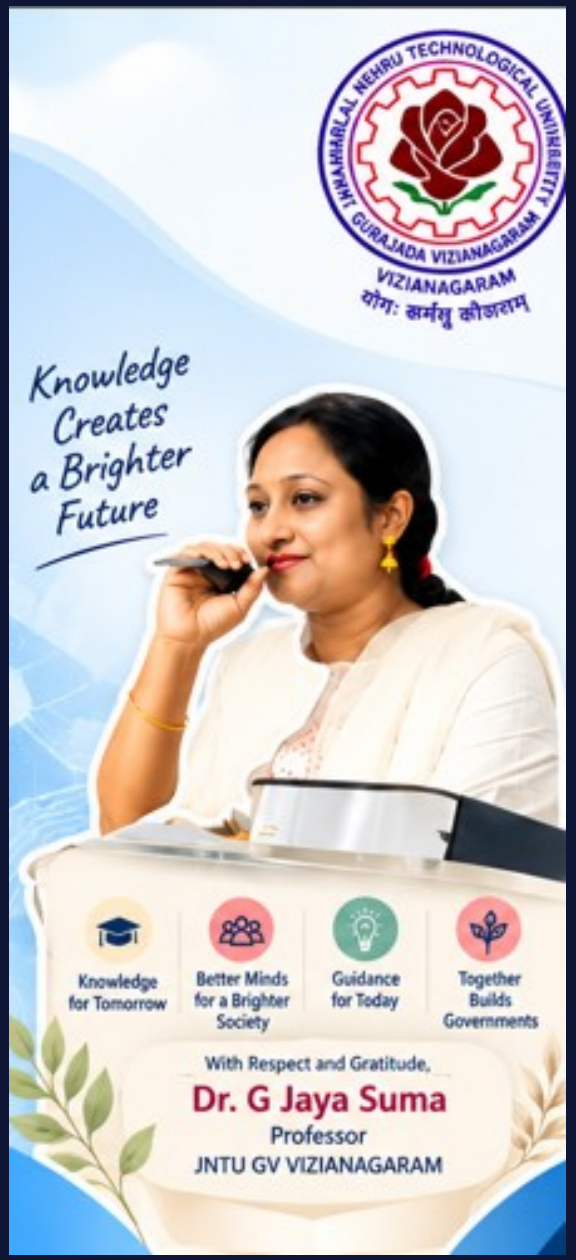
H + T + CNOT works

Clifford alone is not enough

→

Next: Quantum Algorithms

We put universal gate sets to work on real problems, starting with the simplest quantum algorithms.



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VIDEO LECTURE SERIES · CIRCUIT DESIGN

Worked Circuit Designs & Simulation

Design small circuits, predict what they should do, and check the prediction on a simulator.

45 MIN

Design | GHZ state | Half adder | Simulate

Fundamentals of Quantum Computing · Circuit Design & Simulation

1 What we'll cover in 45 minutes

Five segments, from the design loop to debugging with a simulator.

1 Recap: the design-and-simulate loop

Design a circuit, predict its output, simulate, then compare.

5 min

2 Design 1: the GHZ state

Spread one qubit's superposition across three qubits with H and two CNOTs.

10 min

3 Design 2: a reversible half adder

Toffoli and CNOT add two bits — even with an input in superposition.

10 min

4 Predict, simulate, compare

Five circuits checked against their predicted counts.

10 min

5 Debugging and simulation cost

Catching a wrong gate order, and why cost grows with qubit count.

10 min

2 Recap: the design-and-simulate loop

Good circuit design is a loop, not a single step.

In simple words

We design a circuit, predict what it should output, run it on a simulator, and compare. A mismatch means the design or the prediction is wrong.

Predict *vs* **Simulate**

The simulator is a test bench: it checks designs before they ever run on hardware.

1 **Design**

Choose gates and their order to build the state or function you want.

2 **Predict**

Work out the expected counts by hand — before you run anything.

3 **Simulate and compare**

Run many shots and check that the counts match your prediction.

3 Design 1: the GHZ state

One H and two CNOTs spread a superposition across three qubits.

The GHZ design

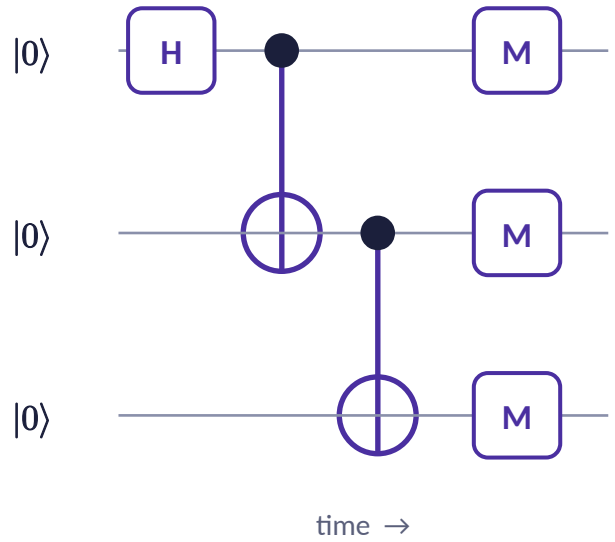
$$|000\rangle \rightarrow 0.707|000\rangle + 0.707|111\rangle$$

H puts the first qubit in superposition; then each **CNOT passes the correlation on** to the next qubit along the chain.

Predict: only 000 and 111 appear, about half each — never a mixed string.

Why GHZ? it is the three-qubit cousin of a Bell pair: all qubits agree, yet each alone looks random.

Circuit for the GHZ state



Read left to right: H on qubit 0, CNOT from 0 to 1, CNOT from 1 to 2, then measure all three.

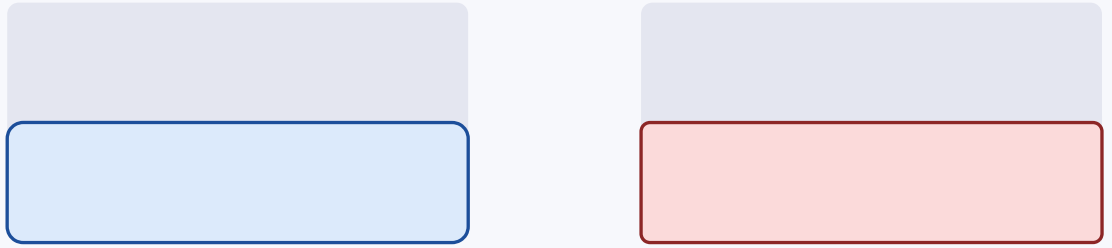
4 Design 2: a reversible half adder

Toffoli and CNOT add two bits — and can do it for both values of a at once.

Example: a in superposition, b = 1

$$|a,\ b,\ c\rangle \rightarrow 0.707|0,1,0\rangle + 0.707|1,0,1\rangle$$

Toffoli writes the carry into the spare qubit c; CNOT turns b into the sum. Both values of a are added at once.



P(0,1,0) = 0.5

P(1,0,1) = 0.5

c

Toffoli: the carry

Flips the spare qubit only when a and b are both 1: carry = a AND b.

s

CNOT: the sum

CNOT from a onto b leaves a $\oplus$ b on b's wire — the sum bit.

!

Order matters

Toffoli must come first: if CNOT ran first, b would already be changed.

5 Predict, simulate, compare

Write the prediction first — then let the simulator confirm or correct it.

Circuit	Prediction	Simulator counts
Bell pair	00 or 11, about half each	{'00': ~512, '11': ~512}
GHZ (3 qubits)	000 or 111, about half each	{'000': ~512, '111': ~512}
Half adder, a = 1, b = 1	Sum 0, carry 1: output 101	{'101': 1024}
Half adder, a = |+⟩, b = 1	010 or 101, half each	{'010': ~512, '101': ~512}
Bug: CNOT first, a = b = 1	Should be 101 — but the order is wrong	{'001': 1024}

6 Debugging with a simulator

The same input gives a different answer when two gates are in the wrong order.

Test input: a = 1, b = 1

$$|a,\ b,\ c\rangle = |1,\ 1,\ 0\rangle$$

A correct half adder gives sum 0 and carry 1. Run the buggy order, CNOT first, on the same input.

↓ *Simulate*

Prediction versus simulated result

101	001
expected (correct design)	observed (CNOT first)

Strings read c b a — Qiskit puts qubit 0 on the right.

Compare with the prediction

A mismatch between counts and prediction points to an error in the design.

Change one thing at a time

Fix the gate order, rerun, and confirm the counts match before adding more gates.

Test small pieces first

Check Toffoli alone, then CNOT alone, before combining them into one circuit.

7 What each design costs to simulate

A simulator tracks 2^n amplitudes, so even simple designs grow fast with qubit count.

Each circuit is listed as qubits · gates

Bell pair	GHZ (3)	SWAP	Half adder	GHZ (10)
Qubits · gates	Qubits · gates	Qubits · gates	Qubits · gates	Qubits · gates
2 · 2	3 · 3	2 · 3	3 · 2	10 · 10
Amplitudes tracked	Amplitudes tracked	Amplitudes tracked	Amplitudes tracked	Amplitudes tracked
4	8	4	8	1,024

The takeaway for this segment: Simulation is ideal for checking small designs, but tracking 2^n amplitudes limits it to roughly 30 to 40 qubits — so test small before scaling up.

ONE-LINE TAKEAWAY

Design, predict, simulate, compare — a small circuit checked on a simulator is a circuit you can trust.

Predict before you run

Test small, then scale

Simulation cost grows as 2^n

→

Next: Quantum Algorithms

We use the same design-and-simulate loop on real algorithms, starting with the simplest ones.